# ECC2K-130: signed σ steps, $R \pm \sigma^j(R)$

`ecc2k130/BEST-OF-TWO.md`, branch `research/ecc2k130-best2-walk-20261014`. Runs on Modal, 2026-10-14: emulation `emulation-v{1,2,3}.jsonl` (108 cells, n = 23–59), device reference walk `device-wc/device-wc-v{1,2}.jsonl` (n = 23, 41), RTX PRO 6000 jobs `gpu-run1` (`SIGMA_SIGN=1`, source `f05102952`) and `gpu-run2` (`SIGMA_SIGN=2`). Iteration counts are exact class-collision counts; GPU rates are medians of interleaved 64-launch samples (A/A then A/B), no isolation receipt.

## Result

The campaign σ walk steps $R \leftarrow R + \sigma^j(R)$. Stepping instead to $R + \sigma^j(R)$ or $R - \sigma^j(R)$ by a class-invariant sign, the parity of popcount$(x \wedge \sigma^j x)$ in normal coordinates (`SIGMA_SIGN=2`), changes only the slope's numerator: no extra field product. Measured:

| quantity | value | source |
|---|---|---|
| rate, RTX PRO 6000, campaign knobs, 385,024 workers | A/B 0.9965 (A/A drift 0.10%); 14.731 ⟶ 14.678 B it/s | `gpu-run2/results` |
| iterations, device walk, n = 41 | 0.958 (device branches), 0.967 (ECC2K-130 probabilities), ±0.012 | `device-wc-v1/v2` |
| iterations, emulation, n = 37–59 | 0.951–0.959 (ECC2K-130 probabilities) | `emulation-v3` |
| correctness | 300/300 replays, 64/64 planted logs, integration + arithmetic pass, 104 regs, 0 spills | `gpu-run2/results` |
| GPU time per solve (derived) | 0.961–0.970 of the σ walk | iterations / rate |

Weight bit 4 as the sign (`SIGMA_SIGN=1`) is rate-neutral (A/B 0.99945) but gains $\leq 1.6\%$ by the model, because the weight is lumpy at n = 131 (minus on 33.9% of steps). No function of the weight can do better. The two walks that **choose** between candidates by a score of the image cut iterations by 13–16%, but they need the second slope: about +38% carry-less work in this kernel (derived, not built).

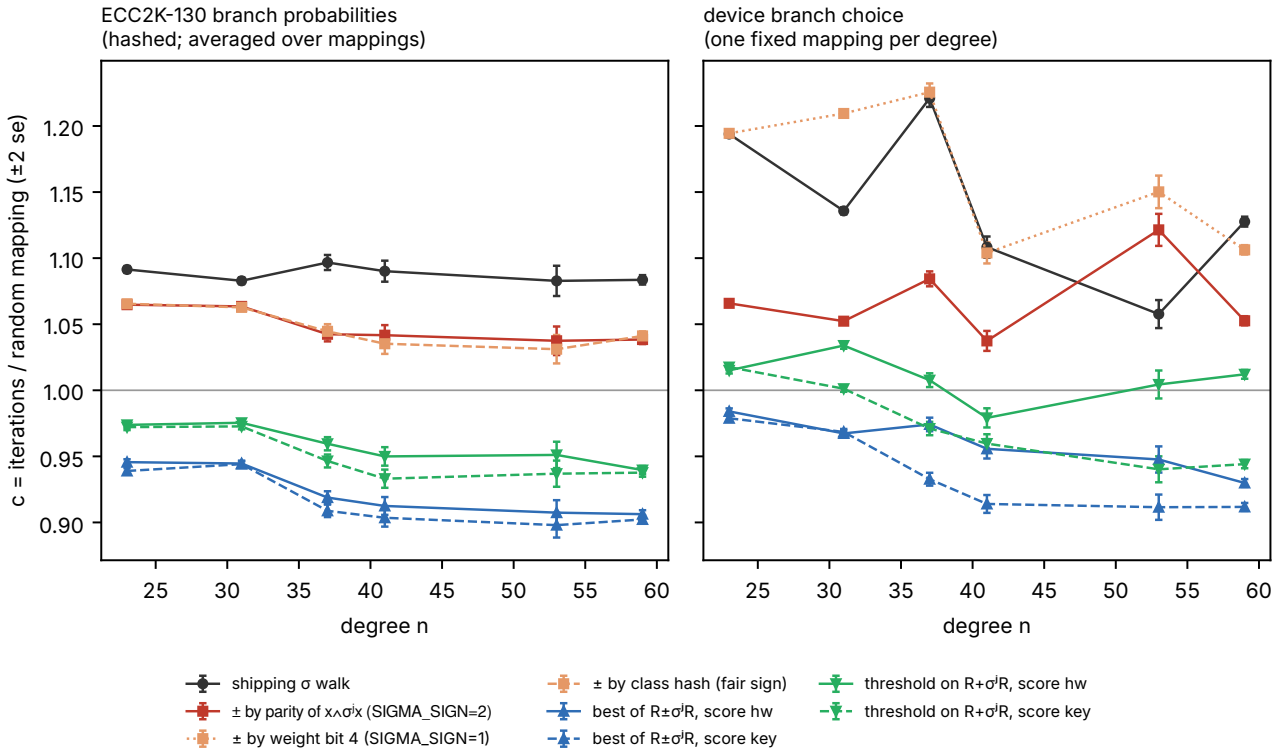



Figure 1: Emulated iteration constant $c$ ($\pm$ 2 se) against degree, every walk. Left: ECC2K-130 branch probabilities, averaged over mappings. Right: the device's branch choice, one fixed mapping per degree; the σ walk itself moves by several percent between mappings.

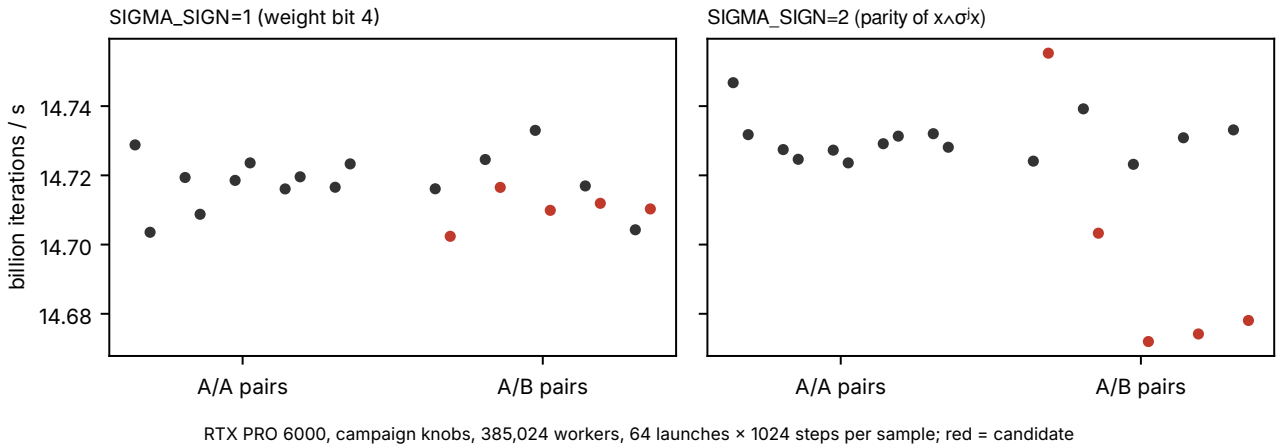



Figure 2: RTX PRO 6000 rate samples, A/A panel then A/B panel (red = candidate), for `SIGMA_SIGN=1` (left) and `=2` (right).

## Caveats

- A different walk: its distinguished points never collide with the current corpus. Switching is the campaign owner's decision, and nothing under the campaign's aws directory is changed.
- The campaign pays for its own n = 131 mapping. Single-mapping rows scatter by several percent, and at n = 53 the signed walk was 6% **worse**. The 3–4% is an average over mappings.
- Short self-returns appear only at $n \le 37$, in the $\sigma$ walk as well, and none is a formal identity in $\mathbb{Z}[\tau]$.
- `WITNESS=1` is refused with the sign (sixteen counters needed); the campaign runs `WITNESS=0`.